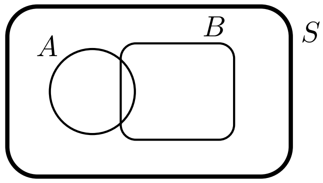
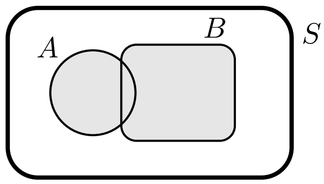
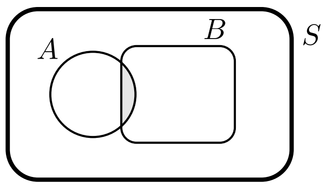
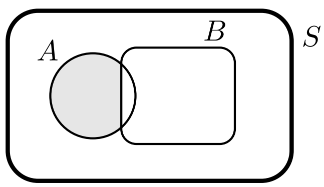
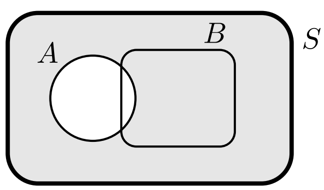
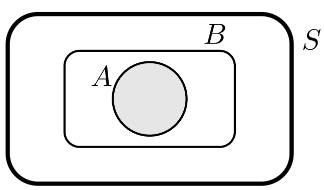
Set Theory, $\boxed{\S}$1, 8

1. Let $A = \{12n \mid n \in \mathbb{Z}\}$, and let $B = \{k \in \mathbb{Z} \mid k \text{ is even}\}$, and let $C = \{3j \mid j \in \mathbb{Z}\}$. Show that $A \subseteq B \cap C$. That is, show that if $x \in A$, then $x \in B$ *and* $x \in C$.

Let's start with some notation and visual intuition. Generally speaking we'll be drawing an example as below, with A and B as subsets of a larger "universe" S.



Operation	**Expression**	**Picture**	**Description/Definition**
Union	$A \cup B$		$x \in A \cup B$ iff $x \in A$ or $x \in B$.
Intersection	$A \cap B$		$x \in A \cap B$ iff $x \in A$ and $x \in B$.
Set difference	$A \smallsetminus B$		$x \in A \smallsetminus B$ iff $x \in A$ and $x \notin B$.
Complement	A' (or A^c or $\overline{A}$)		$x \in A'$ iff $x \notin A$.
Subset	$A \subseteq B$ (or $B \supseteq A$)		$x \in A \implies x \in B$.

For each of the following, determine whether the statement is true for all sets A, B and C. If an equality fails, determine whether one or the other of the possible inclusions holds. Prove your assertions.

2. $(A \cap B) \cup (A \smallsetminus B) = A$

3. $A \cup (B \smallsetminus C) = (A \cup B) \smallsetminus (A \cup C)$

4. Suppose $A = \{k \in \mathbb{Z} \mid 3|k\}$ and $B = \{2k \mid k \in \mathbb{Z}\}$ are subsets of $\mathbb{Z}$. Describe (a) $(A \cup B)'$ and (b) $(A \cap B)'$.

The Joy of Sets – Solutions and Comments

Now, on to the solutions!

1. If $x \in A$, then $x = 12n$ for some integer n. Thus $x = 2(6n)$ (so $x \in B$) and $x = 3(4n)$ (so $x \in C$), and therefore $x \in B \cap C$. That is, $A \subseteq B \cap C$.

2. This seems to be true from a quick sketch (see Figure 1). As usual, to prove two sets are

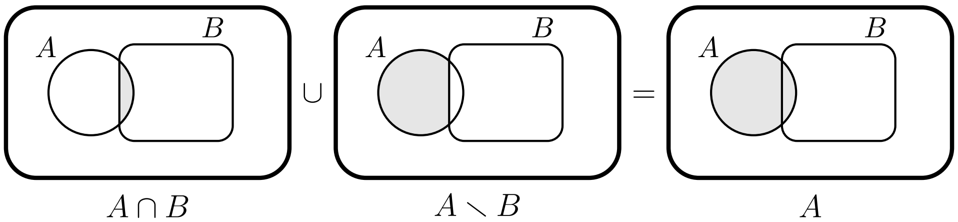


Figure 1: A quick sketch for the answer to Problem 2

equal, we prove each is a subset of the other. That is, to prove $S = T$, we prove that $S \subseteq T$ and $S \supseteq T$. We'll do this in parts:

(a) $(A \cap B) \cup (A \setminus B) \subseteq A$
If $x \in (A \cap B) \cup (A \setminus B)$, then either $x \in A \cap B$ (in which case $x \in A$) or $x \in A \setminus B$ (and again $x \in A$). Thus $x \in A$.

(b) $A \subseteq (A \cap B) \cup (A \setminus B)$
Suppose $x \in A$. If $x \in B$, then $x \in A \cap B$. If $x \notin B$, then $x \in A \setminus B$. Combining these, we see that $x \in (A \cap B) \cup (A \setminus B)$.

3. This is *not true*, since any element of A is included in the set on the left but is not included in the set on the right (it gets "subtracted away" when we remove $A \cup C$ from $A \cup B$). Only one inclusion holds:

Claim: $(A \cup B) \setminus (A \cup C) \subseteq A \cup (B \setminus C)$

Proof. Let $x \in (A \cup B) \setminus (A \cup C)$. Then $x \in A \cup B$ and $x \notin A \cup C$. That is, x is an element of either A or B **AND** x is an element of neither A nor C. That is, $x \notin A$, $x \in B$ and $x \notin C$. Thus $x \in B \setminus C$, so $x \in A \cup (B \setminus C)$. □

4. Here $A \cup B$ can be described as all integers that are multiples of 2 or 3. The complement is therefore all integers that are *not* multiples of 2 or 3. Written out explicitly, we get

$$(A \cup B)' = \{\pm 1\,, \pm 5\,, \pm 7\,, \pm 11\,, \pm 13\,, \pm 17\,, \pm 19\,, \pm 23\,, \pm 25\,, \pm 29\,, \pm 31\,, \pm 35\,, \ldots\}\,.$$

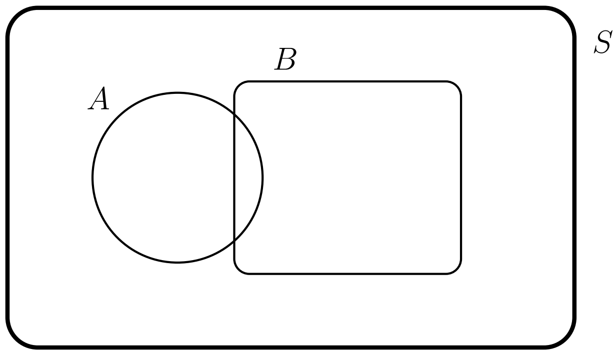
The intersection $A \cap B$ is the set of all integers that are multiples of *both* 2 and 3 – that is, it is the set of integers that are multiples of 6. The complement is thus all integers that are not divisible by 6:

$$(A \cap B)' = \{\pm 1\,, \pm 2\,, \pm 3\,, \pm 4\,, \pm 5\,, \pm 7\,, \pm 8\,, \pm 9\,, \pm 10\,, \pm 11\,, \pm 13\,, \ldots\}\,.$$

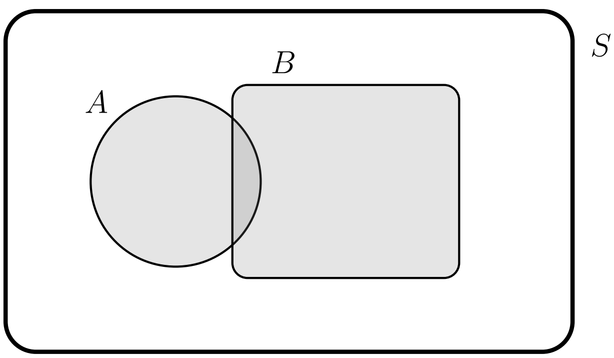
Part of the point of Problem 4 is to introduce the set-theoretic versions of De Morgan's Laws:

$$(A \cup B)' = A' \cap B' \qquad \text{and} \qquad (A \cap B)' = A' \cup B'.$$

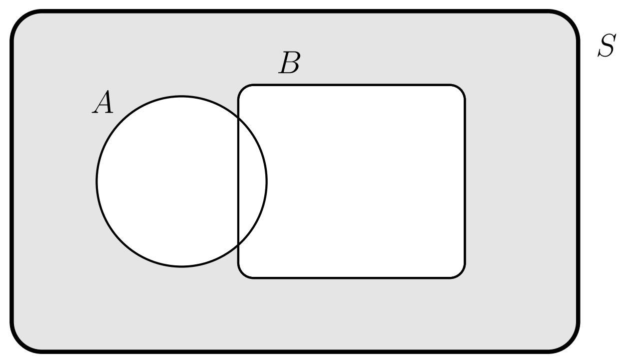
Here are the pictures to illustrate what's going on. Here's our entire set S, with the circular set A and the rectangular set B:



Then here are $A \cup B$ (left) and $A' \cap B'$ (right), showing that $(A \cup B)' = A' \cap B'$:



$A \cup B$ (shaded), so $(A \cup B)'$ is in white



$A' \cap B'$ is shaded

The other law is proved similarly; you can draw the diagram yourself.